Week – 3 Divide and Conquer

Q1. Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int low = 0, high = n-1;
    int mid;
    while(low<high){
        mid = (low + high) / 2;
        if(arr[mid] == 0){
            high = mid-1;
        }
        else if(arr[mid] == 1){
            low = mid+1;
        }
    }
    if(arr[mid]==0){
        if(arr[low]==1)
            printf("%d",n-mid);
        else{
            printf("%d",n-low);
        }
    }
    else
        printf("0");
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Q2. Given an array nums of size n, return *the majority element*.

The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.

Example 1:

Input: nums = [3,2,3]

Output: 3

Example 2:

Input: nums = [2,2,1,1,1,2,2]

Output: 2

Constraints:

- n == nums.length
- $1 <= n <= 5 * 10^4$
- $-2^{31} <= nums[i] <= 2^{31} - 1$

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int majorityElement(int arr[], int n){
    int candidate,count = 0;
    for(int i=0;i<n;i++){
        if(count == 0){
            candidate = arr[i];
        }

        if(candidate == arr[i]){
            count++;
        }
        else{
            count--;
        }
    }

    count = 0;
    for(int i=0;i<n;i++){
        if(arr[i] == candidate){
            count++;
        }
    }

    if(count > n/2){
        return candidate;
    }
    else{
        return -1;
    }
}

int main(){
    int l;
    scanf("%d",&l);
    int arr[l];
    for(int i=0;i<l;i++){
        scanf("%d",&arr[i]);
    }
    printf("%d",majorityElement(arr,l));
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Q3. Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int t;
    scanf("%d",&t);
    int low = 0, high = n-1;
    int mid = (low + high)/2;
    while(low<high){
        if(arr[mid]>t){
            high = mid - 1;
        }
        else if(arr[mid]<t){
            low = mid + 1;
        }
        mid = (high + low) / 2;
    }
    if(arr[mid] < t)
    printf("%d",arr[mid]);
    else
    printf("%d",arr[mid-1]);
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Q4. Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    scanf("%d",arr+i);
    int t;
    scanf("%d",&t);
    int found = 0;
    int a,b;

    for(int i=0;i<n-1;i++){
        a = arr[i];
        b = t - a;
        int low = i+1,high = n;
        int mid = (low + high) / 2;
        while(low<high){
            if(arr[mid] == b){
                found = 1;
                break;
            }
            else if(arr[mid] < b){
                low = mid + 1;
            }
            else if(arr[mid] > b){
                high = mid - 1;
            }
            mid = (low + high) / 2;
        }
        if(found){
            break;
        }
    }
        if(found){
            printf("%d\n%d",a,b);
        }
        else{
            printf("No");
        }
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Q5. Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    scanf("%d",&arr[i]);
    for(int i=0;i<n-1;i++){
        if(arr[i]>arr[i+1]){
            int temp = arr[i];
            arr[i] = arr[i+1];
            arr[i+1] = temp;
            if(i>=1){
                i-=2;
            }
            else if(i==1){
                i--;
            }
        }
    }
    for(int i=0;i<n;i++)
    printf("%d ",arr[i]);
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.